

## LM 5: Server Virtualization

## UNIT I INTRODUCTION TO CLOUD COMPUTING

## INTRODUCTION TO SERVER VIRTUALIZATION:

Server virtualization is a technology that enables multiple virtual instances of servers to run on a single physical machine. By abstracting the underlying hardware, virtualization allows organizations to maximize their computing resources, reduce operational costs, and improve the flexibility of IT infrastructure.

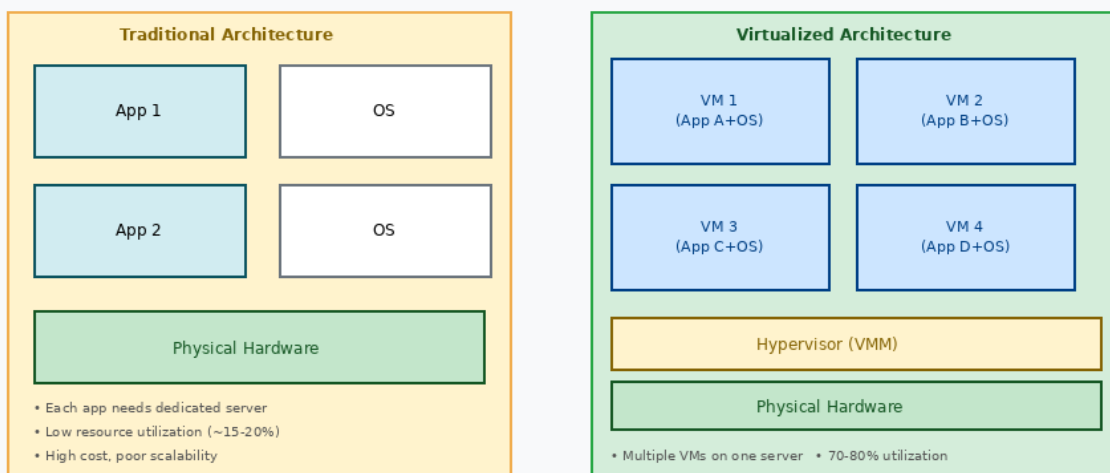
Historically, each application required a dedicated physical server, resulting in underutilized hardware and high operational expenditures. Virtualization revolutionized this model by introducing a software layer that partitions one physical machine into many isolated virtual environments.

## WHAT IS A VIRTUAL MACHINE (VM)?

A Virtual Machine (VM) is a software emulation of a physical computer. It runs an operating system and applications just like a physical computer, but exists only as a file (or set of files) on a physical host. VMs are completely isolated from each other and from the host.

**Key Characteristics of Virtual Machines**

- Complete isolation — each VM is independent and cannot interfere with others
- Encapsulation — the entire state of a VM is stored in files
- Hardware independence — VMs run on any compatible hypervisor
- Portability — VMs can be migrated between physical hosts with minimal disruption
- Snapshot support — the state can be saved and restored at any point

**Traditional vs. Virtualized Architecture**

As shown in Figure 1.1, the traditional model requires one server per application, while virtualization allows multiple applications to share one physical server — dramatically improving efficiency and reducing hardware costs.

### **TYPES OF SERVER VIRTUALIZATION:**

Server virtualization is not a one-size-fits-all solution. There are several distinct approaches, each offering different levels of isolation, performance, and resource sharing. Understanding these types is essential when designing virtualization strategies.

#### **FULL VIRTUALIZATION:**

Full virtualization completely abstracts the physical hardware, creating a fully simulated environment in which an unmodified guest operating system can run. The guest OS is unaware it is running on virtualized hardware.

- Guest OS requires no modification
- Provides complete isolation between VMs
- Slightly lower performance due to binary translation overhead
- Examples: VMware vSphere, Microsoft Hyper-V, Oracle VirtualBox

#### **PARAVIRTUALIZATION:**

In paravirtualization, the guest OS is modified to be aware of the virtualized environment. It communicates directly with the hypervisor through special APIs (hypercalls), resulting in better performance compared to full virtualization.

- Requires modifications to the guest OS kernel
- Higher performance than full virtualization
- Guest OS communicates via hypercalls instead of hardware interrupts
- Examples: Xen paravirtualization, older versions of VMware Tools

#### **OS-LEVEL VIRTUALIZATION (CONTAINERIZATION):**

OS-level virtualization, commonly known as containerization, partitions a single OS kernel to create isolated user-space instances called containers. Unlike traditional VMs, containers share the host OS kernel, making them lightweight and fast.

- Extremely lightweight — no separate OS per container
- Near-native performance since no hypervisor overhead
- Less isolation than full VMs (shared kernel)
- Examples: Docker, LXC, Kubernetes pods, OpenVZ

#### **HARDWARE-ASSISTED VIRTUALIZATION:**

Modern processors include hardware-level support for virtualization. Intel's VT-x (Virtualization Technology) and AMD's AMD-V provide CPU instructions that allow the hypervisor to manage VMs more efficiently, reducing overhead and improving performance.

- Intel VT-x and AMD-V provide CPU-level virtualization support
  - Reduces reliance on binary translation
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- Enables faster context switching between VMs
- Required by most modern Type 1 hypervisors

**HYPERVERSORS — THE CORE OF VIRTUALIZATION**

A hypervisor, also called a Virtual Machine Monitor (VMM), is the software layer that creates and manages virtual machines. It acts as an intermediary between the physical hardware and the virtual machines, allocating and managing resources such as CPU, memory, storage, and network interfaces.

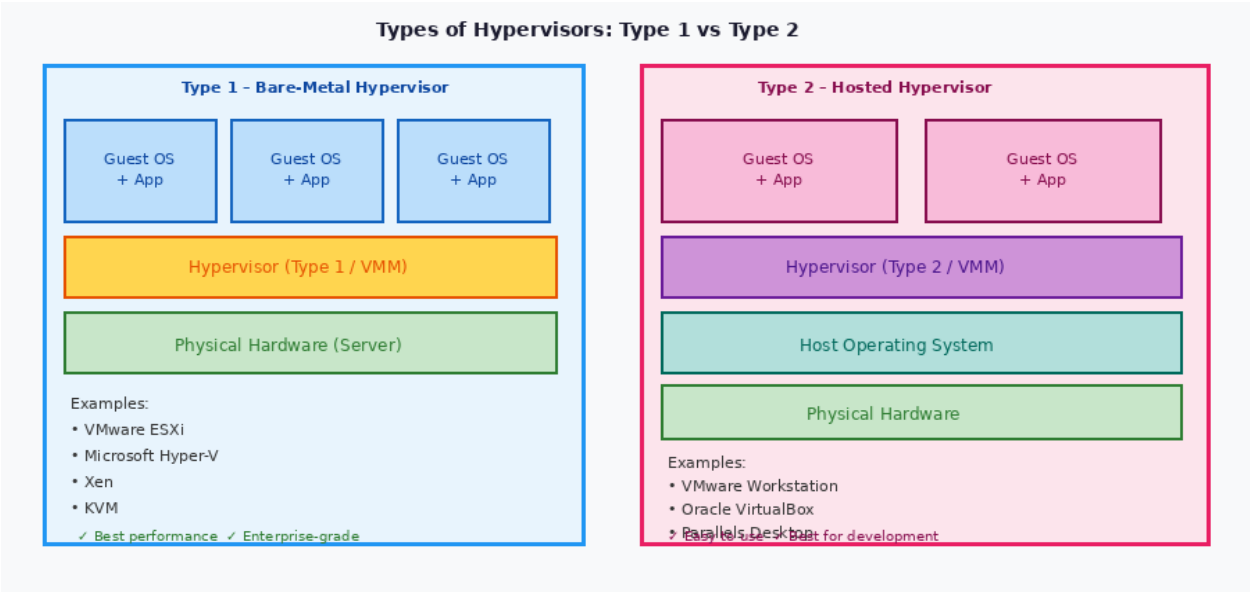


Figure — Type 1 (Bare-Metal) vs. Type 2 (Hosted) Hypervisors

**TYPE 1 HYPERVISOR — BARE-METAL:**

A Type 1 hypervisor runs directly on the physical hardware without a host operating system. It provides the most efficient and highest-performing virtualization environment, making it the standard choice for enterprise data centers and cloud platforms.

| Type 1 Hypervisor — Key Facts                               |
|-------------------------------------------------------------|
| • Runs directly on hardware (no underlying OS needed)       |
| • Provides maximum performance and resource efficiency      |
| • Enterprise-grade security through strong VM isolation     |
| • Supports live migration, HA, and fault tolerance features |
| • VMware ESXi, Microsoft Hyper-V, Citrix XenServer, KVM     |

**Type 2 Hypervisor — Hosted**

A Type 2 hypervisor runs on top of a conventional host operating system. The host OS manages hardware resources, and the hypervisor relies on the OS for I/O management. While less efficient than Type 1, Type 2 hypervisors are ideal for development, testing, and desktop virtualization scenarios.

|                                      |
|--------------------------------------|
| <b>Type 2 Hypervisor — Key Facts</b> |
|--------------------------------------|

|                                                                 |
|-----------------------------------------------------------------|
| • Requires a host OS (e.g., Windows or Linux) to run on         |
| • Easier installation and configuration for end users           |
| • Suitable for development, testing, and educational use        |
| • Lower performance due to extra abstraction layer from host OS |
| • VMware Workstation, Oracle VirtualBox, Parallels Desktop      |

### Comparison: Type 1 vs. Type 2

| Feature     | Type 1 (Bare-Metal)       | Type 2 (Hosted)           |
|-------------|---------------------------|---------------------------|
| Runs on     | Directly on hardware      | Host OS                   |
| Performance | High — minimal overhead   | Moderate — extra OS layer |
| Security    | Strong VM isolation       | Dependent on host OS      |
| Use Case    | Enterprise data centers   | Dev/test, desktop use     |
| Setup       | Complex configuration     | Simple installation       |
| Examples    | VMware ESXi, Hyper-V, KVM | VirtualBox, Workstation   |

### VIRTUAL MACHINE COMPONENTS & ARCHITECTURE:

A virtual machine is a complete software-defined computing environment that mirrors the components of a physical server. Understanding the internal architecture of a VM helps administrators provision, tune, and troubleshoot virtualized workloads effectively.

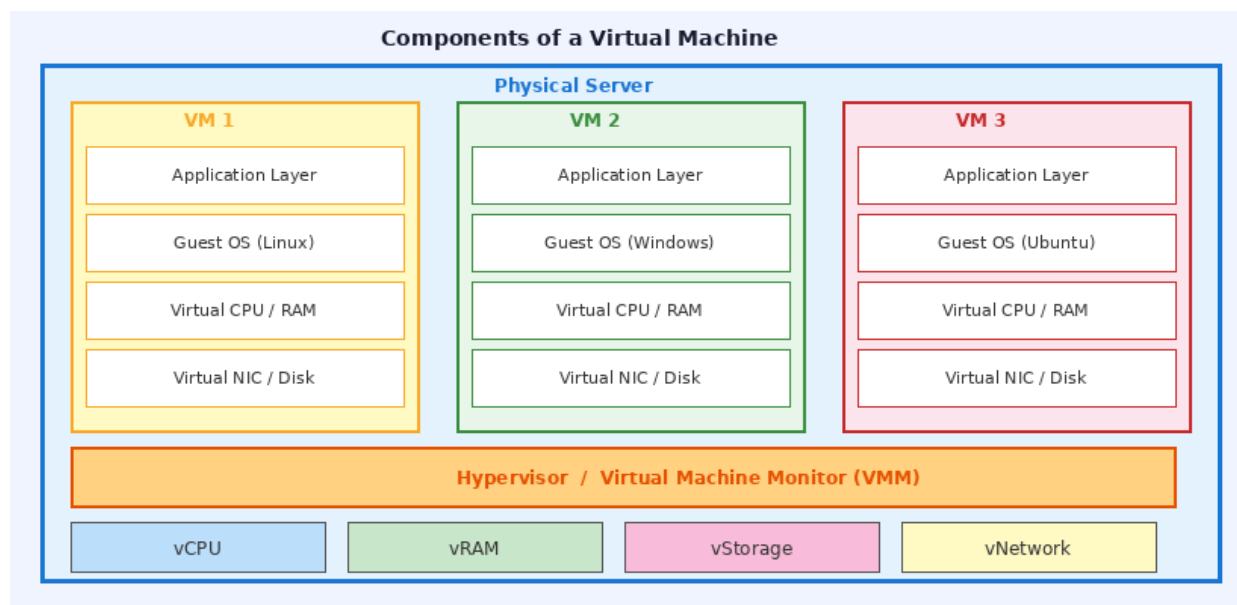


Figure 4.1 — Internal Components of a Virtual Machine

## VIRTUAL CPU (VCPU)

A virtual CPU represents a logical processing unit assigned to a VM. The hypervisor multiplexes physical CPU cores among multiple vCPUs using a scheduler, ensuring fair resource distribution while maintaining performance isolation.

- Each VM can be assigned one or more vCPUs
- The hypervisor scheduler maps vCPUs to physical cores
- Overcommitment allows more vCPUs than physical cores exist
- NUMA-aware placement optimizes memory access patterns

## 4.2 VIRTUAL MEMORY (VRAM)

Virtual memory in VMs operates similarly to physical RAM but is managed by the hypervisor. Techniques like memory ballooning, transparent page sharing, and memory swapping allow hypervisors to manage RAM across multiple VMs efficiently.

- Memory ballooning reclaims idle memory from guest VMs
- Transparent Page Sharing (TPS) deduplicates identical memory pages
- Guest memory can exceed physical RAM through swapping

## 4.3 VIRTUAL DISK & NETWORK

Virtual disks are files (e.g., VMDK, VHD, QCOW2) stored on shared storage or local drives. Virtual network interface cards (vNICs) connect VMs to virtual switches, enabling communication between VMs and external networks.

- VMDK (VMware), VHD/VHDX (Hyper-V), QCOW2 (KVM) are common disk formats
- Thin provisioning allocates disk space on-demand to save storage
- Virtual switches (vSwitch) connect VMs to physical networks
- VLAN tagging enables network segmentation within virtualized environments

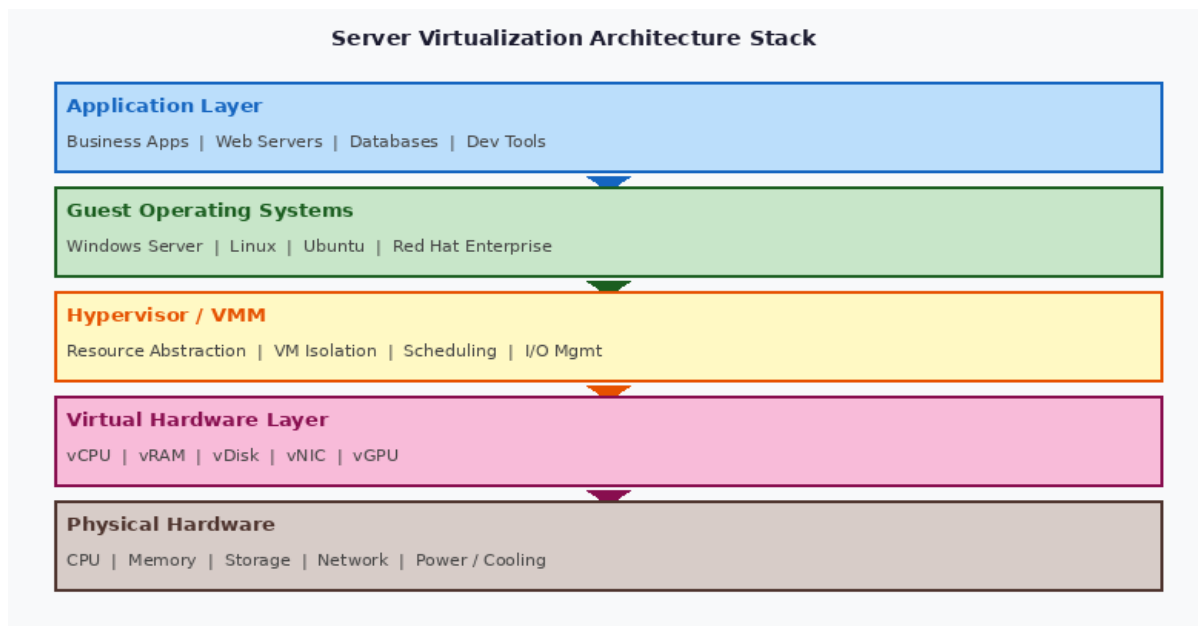


Figure — Server Virtualization Architecture Stack

**BENEFITS & RESOURCE UTILIZATION:**

Server virtualization delivers transformative benefits to organizations of all sizes. By consolidating workloads and abstracting hardware, virtualization leads to significant improvements in efficiency, agility, availability, and cost.

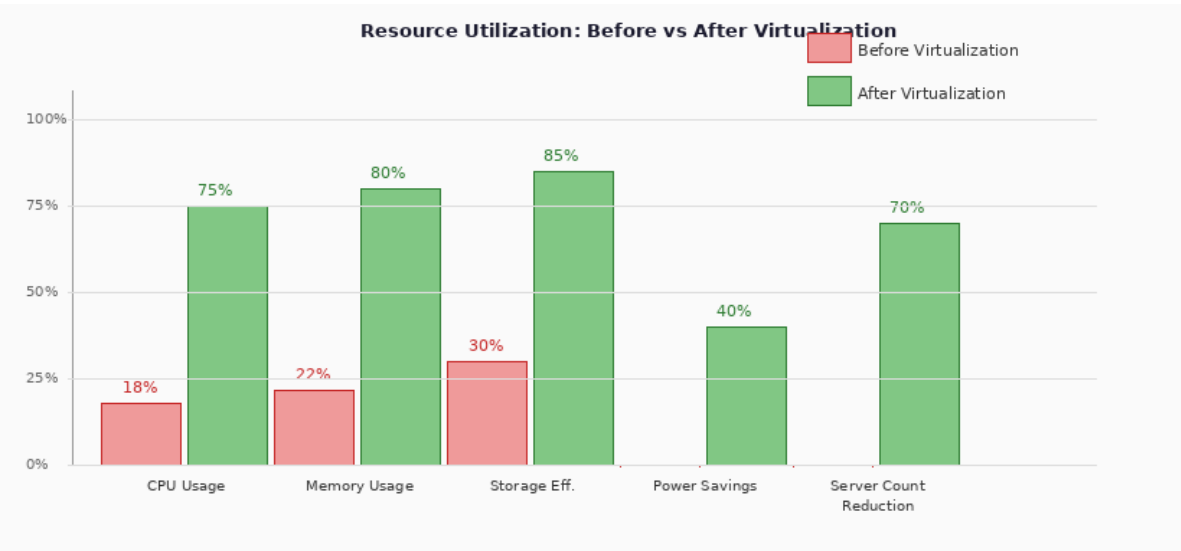


Figure — Resource Utilization Before vs. After Virtualization

**HARDWARE CONSOLIDATION:**

One of the most immediate benefits of virtualization is server consolidation. Organizations routinely achieve consolidation ratios of 10:1 to 20:1, meaning 10–20 physical servers can be replaced with a single well-provisioned host running multiple VMs.

- Average CPU utilization increases from ~18% to 75%+ after virtualization
- Fewer physical servers means lower hardware acquisition costs
- Reduced rack space, cabling, and power distribution overhead
- Simplified hardware maintenance and lifecycle management

**COST SAVINGS:**

Cost reduction occurs at multiple levels: capital expenditure (CapEx) on hardware decreases, and operational expenditure (OpEx) drops due to lower power, cooling, and administration requirements.

| Areas of Cost Reduction Through Virtualization                |
|---------------------------------------------------------------|
| • Hardware costs: Fewer servers needed (10–20x consolidation) |
| • Power consumption: 40–60% reduction in energy bills         |
| • Cooling: Lower heat output reduces HVAC requirements        |
| • Physical space: Smaller data center footprint               |
| • Administration: Fewer physical machines to manage           |
| • Licensing: Streamlined OS and software licensing            |

**HIGH AVAILABILITY & DISASTER RECOVERY:**

Virtualization enables advanced availability features not possible with traditional physical servers. These include live migration, VM snapshots, automated failover, and replication to off-site locations.

- Live Migration (vMotion/Live Migration) — moves running VMs between hosts with zero downtime
- VM Snapshots — capture the VM state at a point in time for backup or rollback
- High Availability (HA) — automatically restarts failed VMs on healthy hosts
- Fault Tolerance — maintains a mirror VM that takes over instantly upon primary failure
- Site Recovery — replicates VMs to a secondary site for disaster recovery

**SERVER CONSOLIDATION & WORKLOAD MANAGEMENT:**

Server consolidation is the strategic process of combining multiple underutilized physical servers onto a smaller number of hosts using virtualization. Effective consolidation requires planning, workload analysis, and ongoing management.

**CONSOLIDATION RATIO PLANNING:**

Determining the optimal consolidation ratio involves profiling current workloads for CPU, memory, storage I/O, and network utilization. Workloads with different peak-usage patterns can be co-located to maximize host utilization.

| Workload Type | CPU Profile    | Memory Need | Consolidation Potential |
|---------------|----------------|-------------|-------------------------|
| Web Server    | Bursty         | Low-Medium  | High (10–15:1)          |
| Database      | High sustained | High        | Low-Medium (3–5:1)      |
| File Server   | Low            | Medium      | High (15–20:1)          |
| Dev/Test VM   | Variable       | Low         | Very High (20+:1)       |
| Email Server  | Moderate       | Medium-High | Medium (5–8:1)          |

**DYNAMIC RESOURCE MANAGEMENT:**

Modern hypervisors include automated resource management capabilities that dynamically balance workloads across hosts in real time. VMware's DRS (Distributed Resource Scheduler) and Microsoft's Dynamic Optimization are leading examples.

- DRS monitors CPU/memory utilization and migrates VMs to balance load
- Storage DRS manages datastore utilization and I/O latency
- Resource pools organize VMs into groups with defined resource shares
- Reservations guarantee minimum resources; limits cap maximum consumption
- Shares determine priority when hosts are under resource contention

**STORAGE & NETWORK VIRTUALIZATION:**

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Virtualization extends beyond compute resources. Storage and network virtualization are equally critical components of a complete virtualization strategy, enabling flexible, scalable, and software-defined infrastructure.

### STORAGE VIRTUALIZATION:

Storage virtualization pools physical storage from multiple devices into a single logical unit that can be allocated to VMs on demand. It decouples storage management from underlying hardware, enabling dynamic provisioning, thin provisioning, and non-disruptive migrations.

- SAN (Storage Area Network) — provides block-level storage over Fibre Channel or iSCSI
- NAS (Network Attached Storage) — file-level storage accessible via NFS or CIFS/SMB
- Software-Defined Storage (SDS) — VMware vSAN, Ceph, Nutanix provide hyperconverged storage
- Thin provisioning allocates only the storage that VMs actively use
- Datastores and LUNs are logical abstractions presented to the hypervisor

### NETWORK VIRTUALIZATION:

Network virtualization abstracts physical network infrastructure into logical networks that can be provisioned, configured, and managed entirely in software. This enables micro-segmentation, flexible topology design, and rapid network changes without touching physical hardware.

- Virtual Switch (vSwitch) — connects VMs to each other and to physical networks
- VLAN Tagging — creates isolated logical networks on shared physical infrastructure
- Software-Defined Networking (SDN) — separates control plane from data plane
- NSX (VMware) and OVN (Open vSwitch) deliver full network virtualization stacks
- Overlay networks (VXLAN, GENEVE) extend Layer 2 networks across Layer 3 boundaries

#### Benefits of Network Virtualization

- Rapid provisioning — deploy new networks in seconds without hardware changes
- Micro-segmentation — apply firewall policies at individual VM level
- Simplified multi-tenancy — each tenant gets isolated virtual network
- East-west traffic visibility — monitor VM-to-VM traffic within the hypervisor

### Virtualization Management Tools:

Managing a virtualized environment at scale requires robust management platforms. These tools provide centralized visibility, automated operations, performance monitoring, capacity planning, and lifecycle management for virtual infrastructure.

### VMWARE VCENTER SERVER:

VMware vCenter Server is the central management platform for VMware vSphere environments. It enables administrators to manage all ESXi hosts and VMs from a single interface, providing access to advanced features such as DRS, HA, vMotion, and storage management.

- Centralized management of hundreds of ESXi hosts and thousands of VMs
  - Role-based access control (RBAC) for granular permission management
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- Performance monitoring with real-time and historical data
- Lifecycle Manager for patch and update management across the environment

**MICROSOFT SYSTEM CENTER (SCVMM):**

System Center Virtual Machine Manager (SCVMM) provides management for Microsoft Hyper-V environments, offering fabric management, VM template libraries, and integration with Azure for hybrid cloud scenarios.

**OPEN-SOURCE TOOLS:**

Several open-source management platforms are widely used in enterprise and research environments:

- oVirt — Red Hat-backed management for KVM virtualization
- Proxmox VE — full-featured management for KVM + LXC containers
- OpenStack — cloud infrastructure orchestration supporting KVM, Xen, and others
- Virt-Manager — graphical management for KVM on Linux desktop environments

**Key Management Operations:**

| Operation              | Description                                                                       |
|------------------------|-----------------------------------------------------------------------------------|
| VM Provisioning        | Create new VMs from templates or ISO images with predefined configurations        |
| Live Migration         | Move running VMs between hosts with zero downtime using vMotion or Live Migration |
| Snapshot Management    | Take point-in-time snapshots of VMs for backup, testing, or rollback              |
| Performance Monitoring | Track CPU, memory, disk I/O, and network utilization in real time                 |
| Capacity Planning      | Forecast resource needs based on historical trends and growth projections         |
| Patch Management       | Apply OS and hypervisor patches to hosts and VMs in a controlled manner           |

**USE CASES, DEPLOYMENT & BEST PRACTICES:**

Server virtualization has been widely adopted across industries and use cases. Understanding real-world applications and following established best practices ensures successful virtualization deployments.

**COMMON USE CASES:**

| Use Case                  | Details                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------|
| Data Center Consolidation | Replace hundreds of underutilized physical servers with a smaller number of high-density hosts   |
| Cloud Computing           | Public, private, and hybrid clouds are built on virtualization (AWS EC2, Azure VMs, GCP Compute) |

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|                              |                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------|
| Development & Testing        | Quickly spin up isolated VMs for testing without impacting production environments  |
| Desktop Virtualization (VDI) | Deliver virtual desktops from data center to end-users via VMware Horizon or Citrix |
| Disaster Recovery            | Replicate VMs to secondary sites and fail over rapidly during outages               |
| Legacy App Hosting           | Run legacy OS and applications on modern hardware using VMs                         |
| Security Sandboxing          | Execute untrusted code in isolated VMs to prevent host compromise                   |

### DEPLOYMENT BEST PRACTICES:

A successful virtualization deployment requires careful planning, standardized processes, and ongoing governance. The following best practices are recommended by industry leaders and virtualization vendors:

1. **Assess workloads before migration** — Profile CPU, memory, storage, and network usage to right-size VMs.
2. **Use VM templates** — Create standardized templates for common workloads to ensure consistency and speed up deployment.
3. **Implement resource governance** — Use resource reservations, limits, and shares to prevent noisy-neighbor issues.
4. **Plan for storage performance** — Choose appropriate storage tiers (SSD vs HDD) based on workload I/O requirements.
5. **Segregate network traffic** — Use dedicated NICs/VLANs for management, storage, vMotion, and VM traffic.
6. **Implement regular backup and DR testing** — Test VM backups and disaster recovery procedures at least quarterly.
7. **Monitor continuously** — Deploy monitoring tools to track utilization, performance, and capacity trends.

### VIRTUALIZATION VS. CONTAINERIZATION:

Virtualization and containerization (e.g., Docker, Kubernetes) are complementary technologies. While VMs provide strong isolation and full OS environments, containers are lightweight and suited for microservices architectures. Many modern deployments run containers inside VMs for the best of both worlds.

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